

Al-Mehedi Hasan, Md. Mehedi Hasan Chowdhury Mahi, Tanver Ahmed, Md. Muktar Hossain
Department of Computer Science and Engineering, Varendra University

ABSTRACT

Fraud in credit cards continues to cost the world huge sums of money and this has prompted the invention of automated and data-driven systems to detect fraud. Machine learning systems have demonstrated high performance in identifying abnormal transaction behavior, especially on highly imbalanced datasets. This paper presents a comparative analysis of ten machine learning classifiers using SMOTE balancing on a real credit card fraud dataset. The models are evaluated using accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, and confusion matrix-based metrics. The results show that ensemble-based models, particularly Histogram-Based Gradient Boosting and XGBoost, outperform linear and probabilistic classifiers.

Keywords:

Credit Card Fraud Detection, Machine Learning, Ensemble Methods, Imbalanced Data, Comparative Evaluation, SMOTE.

INTRODUCTION

- Online payment growth has led to a rapid increase in credit card fraud.
- Traditional rule-based systems fail to adapt to evolving fraud patterns.
- Machine learning techniques offer adaptive and data-driven solutions.
- Severe class imbalance makes fraud detection challenging.
- This study compares ten ML classifiers using SMOTE for effective fraud detection.

OBJECTIVE

To compare ten machine learning classifiers for credit card fraud detection on a highly imbalanced dataset using SMOTE and unified evaluation metrics to identify the most effective models.

METHODOLOGY

Dataset:

- Publicly available Kaggle credit card transaction dataset
- Highly imbalanced:
 - Legitimate transactions: 6,341,907
 - Fraudulent transactions: 7,717
- Features anonymized using PCA for privacy protection

Preprocessing:

- Missing value handling and normalization
- SMOTE applied to oversample minority (fraud) class
- Balanced training data used for all models

Models Implemented:

- Histogram-Based Gradient Boosting
- XGBoost
- Extra Trees
- Random Forest
- Gradient Boosting
- Multi-Layer Perceptron
- K-Nearest Neighbors
- Logistic Regression
- AdaBoost
- Gaussian Naive Bayes

RESULTS AND DISCUSSION

- Accuracy alone is misleading due to class imbalance
- Ensemble models outperform other classifiers across F1-score, PR-AUC, and ROC-AUC
- **Best Performers:**
 - **HistGB:** F1 = 0.878, ROC-AUC $\approx$ 0.9998
 - **XGBoost:** F1 = 0.871, high precision and low false positives
- Traditional and probabilistic models show high false positives and poor balance between recall and precision
- Gaussian Naive Bayes performs worst due to strong independence assumptions

Table I. Comparison of machine learning techniques based on accuracy

Model	Accuracy	F1	Precision	Recall	PR_AUC	ROC_AUC	TN	FP
HistGB	99.93%	0.878179	0.930336	0.831559	0.957941	0.999787	1266247	234
XGB	99.93%	0.870801	0.945449	0.807078	0.950435	0.999774	1266306	175
ExtraTrees	99.92%	0.850214	0.956739	0.765035	0.9188	0.997762	1266351	130
RF	99.91%	0.827168	0.96124	0.725918	0.918198	0.999555	1266371	110
GradBoost	99.92%	0.852924	0.970468	0.760777	0.896111	0.999105	1266394	87
MLP	98.65%	0.297044	0.175482	0.966738	0.788512	0.997115	1249411	17070
KNN	99.87%	0.746492	0.901355	0.637041	0.71421	0.900189	1266219	262
LogReg	99.83%	0.661241	0.781772	0.572911	0.696413	0.988725	1265880	601
AdaBoost	99.15%	0.297391	0.197139	0.605109	0.150356	0.976719	1257220	9261
GNB	59.30%	0.014054	0.007078	0.980575	0.046509	0.874491	749525	516956

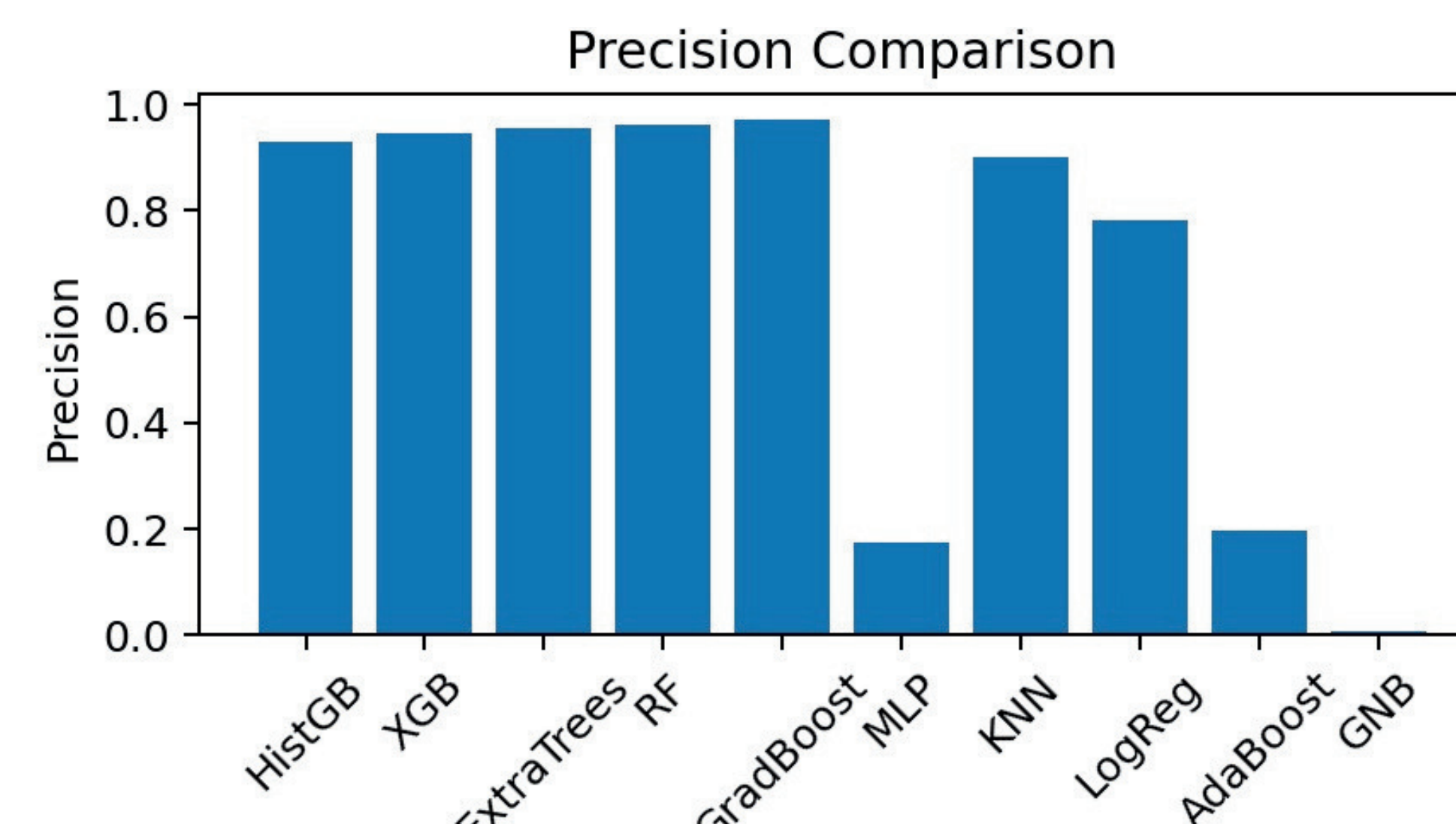


Fig. 1. Comparison of machine learning models based on precision

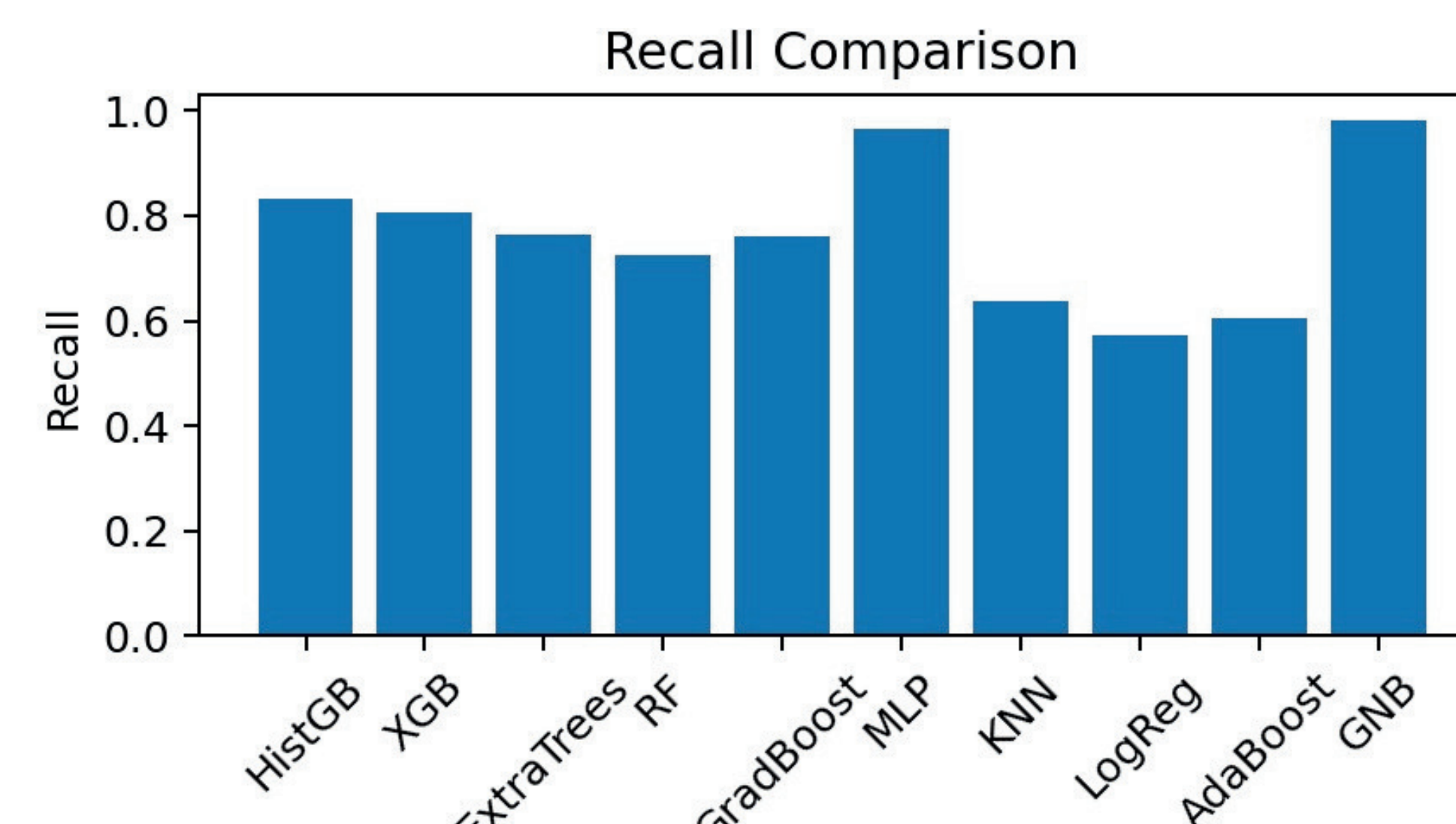


Fig. 2. Comparison of machine learning models based on recall

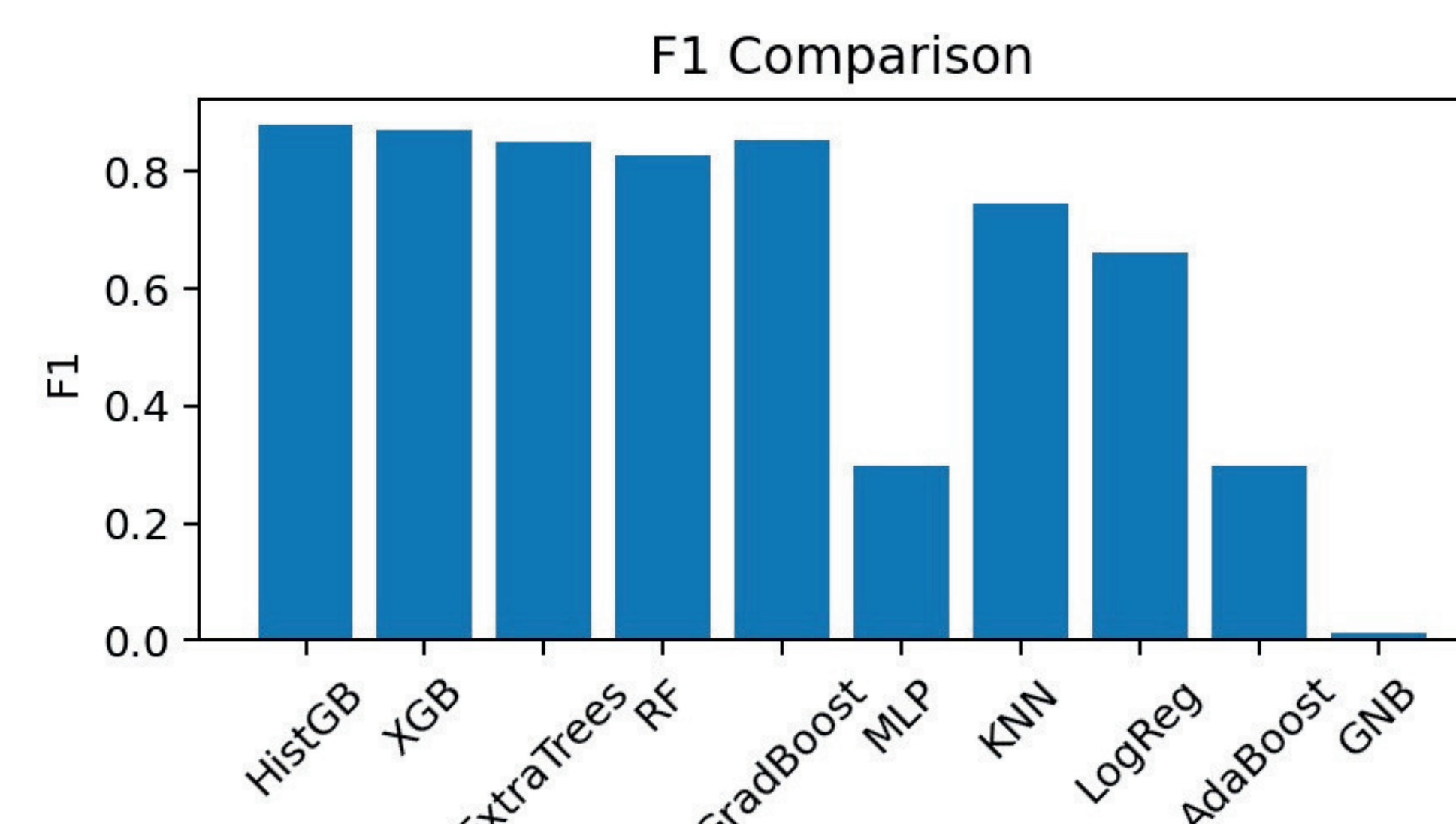


Fig. 3. Comparison of machine learning models based on F1-score

CONCLUSION

- SMOTE effectively improved fraud detection on highly imbalanced data.
- Tree-based ensemble models outperformed all other classifiers.
- Histogram-Based Gradient Boosting and XGBoost achieved the best overall performance.
- Linear and probabilistic models showed higher false positives and lower reliability.
- Ensemble learning is essential for robust credit card fraud detection.

REFERENCES

- [1] Alarfaj, F. K., Malik, I., Khan, H. U., Almusallam, N., Ramzan, M., & Ahmed, M. (2022). Credit card fraud detection using state-of-the-art machine learning and deep learning algorithms. *IEEE Access*, 10, 39700-39715.
- [2] Salunke, Y., Phalke, S., Madavi, M., Kumre, P., Bobhate, G., Madavi, M. D., & Kumre, P. D. (2025). Fraud Detection: A Hybrid Approach With Logistic Regression, Decision Tree, and Random Forest. *Cureus Journals*, 2(1).
- [3] Bhakta, S. S., Ghosh, S., & Sadhukhan, B. (2023, August). Credit card fraud detection using machine learning: A comparative study of ensemble learning algorithms. In *2023 9th International conference on smart computing and communications (ICSCC)* (pp. 296-301). IEEE.